Definition (Associative stacking semicategory)
An *associative stacking semicategory* is a stacking semicategory $\langle \mathbf{C}, \otimes, \otimes \rangle$ satisfying the following.

<u>Conditions</u>

- The stacking operation $\otimes$ makes the objects into a semigroup.
- The stacking operation $\otimes$ makes the morphisms into a semigroup.